Specification: A Continuous-Quote RFQ Market for Perpetual Options on a Burr-II Kernel, with its Formal (Lean 4) Backing

Temporal Exchange working notes*

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Abstract

We specify a request-for-quote (RFQ) market-maker console for perpetual options in which a maker's quote at *every* strike is a single closed-form curve: four parameters $(\bar{S}, a, \gamma, \kappa)$ of a Burr-II (Burr Type XII in the moneyness modulus) wing kernel, plus a half-spread h. There is no order ladder: any strike is priced on demand, requoting is four parameters rather than N orders, and the two curve sets (bid and ask) make the spread an irreversible revenue band. Makers differ only in depth and spread while sharing the price *level* — a design choice motivated by a proved theorem (heterogeneous per-maker steepness generates a strict volatility smile that no single power-law lens can represent; on the Burr-II kernel itself the mixture effect is proved second-order, so there the common level is chosen, not forced). The tradeable book is the envelope of maker curves; RFQ fills walk makers cheapest-first, each paid at its own posted spread; settlement nets in perpetual units and crosses one common units→cash doorway, which is exactly the arbitrage-free class. We give the closed form, the quoting economics (vol-indexed spreads against gamma bleed), the aggregation and fill rules, the user-interface contract, worked numbers re-derived from the shipped code, and the complete formal corpus backing the design: five Lean 4 packages, 130 theorems, all kernel-checked locally with axiom set $\{\mathsf{propext}, \mathsf{Classical.choice}, \mathsf{Quot.sound}\}$. Open gaps are stated rather than hidden.

1 Scope and provenance

Object specified. `app/index.html` (a single 562-line file; Burr-II math in JavaScript including a continued-fraction regularized incomplete beta), `app/server.js`, `app/railway.json`. Build 11 = commit `ebd774e` (2026-08-14); deployed at `https://web-production-cc265.up.railway.app`.

Status. The console is a *brainstorm simulation* (its own footer says so): it is not the production engine (`engine/builds/HEAD_temporal_mvp_v28_lens.html`, which is untouched by all of this), and it uses demo constants — spot $S = \$65{,}695.5$, three synthetic competitor makers, demo portfolio positions.

Verification posture. Every number in §6 was re-derived on 2026-08-18 by executing the shipped code's own mathematics in Node.js, not copied from build logs. The Lean corpus

*Specification of the market-maker console `app/index.html`, builds 1–11 (2026-08-14), repository `Perp-Options-AMM`. Prepared 2026-08-18. This document specifies a *brainstorm simulation*, not the production engine; see §1.

of §7 was proved by the external prover Aristotle (Harmonic) or written in-repo, then *locally kernel-verified* in our own Lean 4.28.0 + Mathlib kernel; caveat stated, not hidden: Mathlib's `.olean`s came from the Mathlib CI cache, the universal practice, not a from-scratch rebuild. The JavaScript↔Lean correspondence is statement-by-statement mirroring plus numeric spot checks; the transcription itself is asserted, not proved.

2 The pricing kernel

Let S be spot, K a strike, and $k = K/S - 1$ the linear moneyness (calls are out of the money for $k \geq 0$, puts for $k \leq 0$; the zero-strike edge is $k = -1$). Fix parameters $\bar{S} > 0$ (mean strike scale), $a > 0$ (Burr shape), $\gamma > 0$ (tail weight), $\kappa \in (-1, 1)$ (skew). Write $B = \Gamma(\frac{1}{a})\Gamma(\frac{\gamma}{a})/\Gamma(\frac{1}{a} + \frac{\gamma}{a}) = \mathrm{B}(\frac{1}{a}, \frac{\gamma}{a})$ and let $I_u(\cdot, \cdot)$ be the regularized incomplete beta function.

Wing scales and kernel.

$$s_R = \bar{S}(1+\kappa), \qquad s_L = \bar{S}(1-\kappa), \qquad \Delta_s(v) = \Big(1 + (v/s)^a\Big)^{-\frac{\gamma+1}{a}}. \tag{1}$$

Wing value (tail integral), pedestal, masses.

$$T_s(m) = \frac{s}{a}\, B\left(1 - I_u(\tfrac{1}{a}, \tfrac{\gamma}{a})\right), \quad u = \frac{m^a}{m^a + s^a}, \qquad G_1 = \Delta_{s_L}(1), \quad I_1 = T_{s_L}(1), \tag{2}$$

$$W_R = \frac{s_R B}{a}, \qquad W_L = \frac{s_L B/a - I_1 - G_1}{1 - G_1}. \tag{3}$$

The peg (derived, not chosen).

$$q_R = \frac{W_L}{W_R + W_L}, \qquad q_L = 1 - q_R \tag{4}$$

is the *unique* split making the two wings meet at the money (Lean: `atm_wings_meet`); at-the-money put–call parity *is* the peg.

Wings and option values.

$$A_R(k) = q_R\, T_{s_R}(|k|), \tag{5}$$

$$A_L(k) = \begin{cases} \dfrac{q_L}{1-G_1}\Big(T_{s_L}(|k|) - I_1 - G_1\,(1-|k|)\Big), & |k| < 1, \\ 0, & |k| \geq 1, \end{cases} \tag{6}$$

$$\mathrm{CALL}(k) = \begin{cases} A_R(k), & k \geq 0 \\ -k + A_L(k), & k < 0 \end{cases} \qquad \mathrm{PUT}(k) = \begin{cases} A_L(k), & k \leq 0 \\ k + A_R(k), & k > 0. \end{cases} \tag{7}$$

Out of the money the value is the wing; in the money it is intrinsic plus the *mirror* wing, so

$$\boxed{\mathrm{CALL}(k) - \mathrm{PUT}(k) = -k \quad \text{at every strike}} \tag{parity}$$

(Lean: `burr2_parity`; away from $k = 0$ it is hypothesis-free branch bookkeeping, `burr2_parity_of_ne`). The left wing's closed form already vanishes at the zero-strike edge (`AL_glue`), giving the exact anchors $\mathrm{CALL}(-1) = 1$ (the whole coin) and $\mathrm{PUT}(-1) = 0$. The at-the-money mark is $\mathrm{ATM} = q_R W_R$. Sign and order — all values nonnegative, CALL antitone and Lipschitz, PUT monotone — hold conditionally on a two-sided Riemann-sandwich interface for the incomplete beta (`TailRep`, proved non-vacuous by an explicit witness); see §8.

3 Quoting: two curve sets and vol-indexed spreads

The curve is the quote. A maker posts $(\bar{S}, a, \gamma, \kappa)$ and a half-spread h (bps). At any strike, on demand,

$$\mathrm{ask}(k) = V(k)\,\big(1 + \tfrac{h}{10^4}\big), \qquad \mathrm{bid}(k) = V(k)\,\big(1 - \tfrac{h}{10^4}\big), \tag{8}$$

with $V \in \{\mathrm{CALL}, \mathrm{PUT}\}$. There is no ladder to post, cancel, or replace; requoting is four parameters, not N orders. The band between the two curve sets is the maker's revenue and is *irreversible* (unlike constant-product-AMM slippage, which a round trip refunds). Execution is priced on the *continuous (integral) convention*: impact is a price path, not maker revenue — the endpoint-price shortcut that fabricated "slippage revenue" was identified and removed before this app was built (operator rulings, transcript entries 539–540).

Gamma bleed and the fair spread. With cash value $C_{\text{cash}}(S, K) = \mathrm{CALL}(K/S - 1) \cdot S$, the console measures a per-dollar-notional cash curvature

$$\tilde{G} = \operatorname*{mean}_{K/S \in \{0.85, 0.95, 1, 1.05, 1.15\}} \Big| S\, \partial_S^2 C_{\text{cash}}(S, K) \Big|, \qquad \mathrm{bleed} = \tfrac{1}{2}\, \tilde{G}\, \mathrm{RV}^2 \;\; [\text{notional/yr}], \tag{9}$$

and the fair half-spread and quote

$$h_{\text{fair}} = \frac{\mathrm{bleed}}{\mathrm{turnover} \cdot 365} \times 10^4 \text{ bps}, \qquad h_{\text{eff}} = \begin{cases} h_{\text{fair}} \cdot (1 + 25\%) & \text{vol-indexed mode,} \\ \mathrm{fee} \cdot 10^4/2 & \text{fixed-spread mode.} \end{cases} \tag{10}$$

Fees, bleed and returns per year on quoted notional $N_\$$ (margin $\times$ 10 pool leverage, both wings) are

$$\mathrm{fees} = \tfrac{h_{\text{eff}}}{10^4}\, \mathrm{turn} \cdot 365 \cdot N_\$, \quad \mathrm{bleed}_\$ = \mathrm{bleed} \cdot N_\$, \quad \mathrm{APR}_{\text{base}} = \tfrac{\mathrm{fees} - \mathrm{bleed}_\$}{N_\$}, \quad \mathrm{APR}_{\text{lev}} = \tfrac{\mathrm{fees} - \mathrm{bleed}_\$}{\mathrm{equity}}. \tag{11}$$

Capacity is a *continuous* density $w(k) \propto e^{-8\lambda k^2}$ normalized to the wing notional — the ribbon in the chart, not rungs. The economics layer is a model with no Lean backing (§8).

4 Aggregation, the RFQ, and settlement

Common level, private transport. All makers share the level parameters $(\bar{S}, a, \gamma, \kappa)$; each maker owns only its depth λ_i and half-spread h_i. The design is *chosen, motivated by* — not forced by — a proved theorem: `mixture_not_single_lens` (§7) proves, *for the power-law lens family*, that mixing distinct per-maker steepnesses produces a strictly log-convex (smiled) aggregate no single lens represents. On the Burr-II kernel itself the corpus's own `BURR2_MIXTURE` result shows the mixture effect is second-order mild and re-fit is practically viable (transcript entry

529), and the envelope need not be one curve — so common level is a choice here. Its ratification provenance is reconstruction-grade (the 2026-08-14 session is untranscribed; "transport-not-level" enters the transcript as a recommendation, entry 513). Build 11 exhibited a related but *distinct* failure empirically: per-maker levels crossed the book (best bid 0.15482 > best ask 0.12015; −1,261 bps in half-spread convention, −2,522 bps full-spread) — a level-disagreement failure, not an instance of the smile obstruction. What is provable: under a common level and positive spreads the book cannot cross. The shipped design is "transport, not level," with an honest CROSSED — arb detector.

Depth aggregation (harmonic).

$$\frac{1}{\lambda_{\text{agg}}} = \sum_i \frac{1}{\lambda_i}, \qquad \text{share}_i = \frac{1/\lambda_i}{\sum_j 1/\lambda_j} \tag{12}$$

(Lean: `harmonic_law`, `shares_sum_one`, `share_eq_agg_ratio`; the parallel-addition law itself is *earned* in the base layer from walking a stacked ladder to a common marginal price, `walk_equiv`/`walk_cost_equiv`, interior case).

Envelope book and RFQ fill.

$$\text{ask}(k) = \min_i V_i(k)\big(1 + \tfrac{h_i}{10^4}\big), \qquad \text{bid}(k) = \max_i V_i(k)\big(1 - \tfrac{h_i}{10^4}\big), \tag{13}$$

so the aggregate is always at least as tight as any single maker (the "improvement vs. single maker" panel quantifies what competition buys). An RFQ of size Q walks makers cheapest-first, fills each up to $\text{share}_i \times \text{pool}$, and pays each maker *at its own posted spread* (pay-your-own-quote, not uniform-price; Lean: `apportionment_conserves`).

Settlement doorway. Legs net in perpetual *units*; cash appears only through one common conversion at the exit. This is forced, not convenient: `doorway_arbfree_iff_common` proves a book netting to zero units pays zero cash *iff* the units→cash factor is common across strikes, and any per-strike factor admits unbounded arbitrage (witness: long $1/V_i$ at strike i, short $1/V_j$ at strike j — zero net units, cash $F_i - F_j$, scalable). `exit_timing_irrelevant`: a netted-flat book settles to zero for every closing equity and leverage.

Trade bands and portfolio. A band sells one wing and buys the other with the proceeds (per-leg fee and slippage); *only κ moves* — level and shape stay put — and every strike re-prices at once. The portfolio view keeps perps and perp-options in one account equity (account-level liquidation, 50× cap with headroom), carves option-backing notional out of the perps tab, and reads net option delta back to a perp hedge. The displayed "residual exposure 0.000000" is a hardcoded string over the definitional identity hedge $:= -\Delta_{\text{net}}$ — closure by construction, not a computed check; a wrong Δ would still "read back" to zero, and the demo put-leg delta is in fact not parity-consistent (skeptic finding, §8 item 5). The κ dynamics and the exposure readback have *no* Lean backing (§8).

5 User-interface contract

view	contract
Earn (maker)	Curve parameters $\bar{S} \in [.05, 1]$, $a \in [.2, 3]$, $\gamma \in [.3, 4]$, $\kappa \in [-.9, .9]$, $\lambda \in [.01, 1]$, fee $\in [0, .2]$, depth $\in [1, 80]$ BTC/1%-strike; margin. Chart: call/put ask solid, bid dashed, both first-class; spread drawn as a filled band; "spread view $\times N$" magnifier (default $\times 25$) with an honest caption — at $\sim$19 bps the true gap would overlap on screen; mid demoted to faint reference; capacity-density ribbon. *Hover quoting*: a crosshair tracks any strike; readout = strike, call bid/ask, put bid/ask, half-spread, $C - P$ vs $-k$ parity, spread capture on size. No standalone quote box (the chart *is* every strike). Market sliders: realised vol, turnover/day, RFQ size. Risk: curvature, break-even half-spread and turnover, parity check to 12 dp. Vol-indexed toggle.
Transact (taker)	Buy/sell, any strike $k \in [-60\%, +60\%]$, size. Envelope best-ask solid / best-bid dashed with individual makers faint behind; order ticket prices off the envelope and names the filling maker; fill table walks makers cheapest-first; book state flag clean / CROSSED — arb.
Trade Bands	Sell-strike, buy-strike, size; both leg prices, proceeds, units bought; state move $\kappa \to \kappa + \Delta\kappa$ with $\bar{S}, a, \gamma$ explicitly unchanged; curve drawn before/after.
Portfolio	Positions with marks and deltas; one account equity, leverage vs the 50× cap with headroom and SAFE/LIQUIDATED; carve panel; hedge readback (net option $\Delta \to$ required perp hedge $\to$ residual 0).

6 Worked numbers (re-derived from the shipped code)

Defaults: $\bar{S} = .60$, $a = 1.2705$, $\gamma = 1.8413$, $\kappa = 0$, $\lambda = .10$, fee $= .02$, margin 15 BTC, $S = \$65{,}695.5$, RV 60%, turnover 0.30×/day, pool 200 BTC, RFQ 5 BTC. All values recomputed 2026-08-18 by executing the app's own math in Node.

quantity	value	note
ATM mark	0.178888	$q_R W_R$
CALL/PUT at k=+12%	0.135450 / 0.255450	Transact default strike
parity $\lvert C - P + k \rvert$, k=0.2	0	exact in floating point off-ATM
parity $C - P$ at k=0	2.44×10^{-13}	the peg, numerically (continued-fraction I
CALL(−1) / PUT(−1)	1 / 0	zero-strike anchor, exact
curvature $\tilde{G}$	0.9065	five-point mean
bleed	0.16317/yr	fraction of notional
h_{fair} / h_{eff}	14.90 / 18.63 bps	fair; vol-indexed +25%
Σ notional	300 BTC = $19,708,650	margin ×10, both wings
fees / bleed per yr	$4,019,813 / $3,215,850	
base / levered APR	4.1% / 81.6%	levered on 15 BTC equity
break-even turnover	0.24×/day	vs 0.30× assumed
envelope at k=12%	ask 0.13567 / bid 0.13523	book spread 32.8 bps (best-maker half-spr
depth shares	47.6/19.0/9.5/23.8 %	you/Kappa/Delta/Sigma, harmonic
band (sell +10% 5 BTC, buy −10%)	proceeds 0.6840 BTC, buys 5.2346 u	$\Delta\kappa = +0.00574$; $\bar{S}, a, \gamma$ unchanged

Vol sweep (levered APR, current build-11 code).

realised vol	h_{fair} (bps)	vol-indexed	fixed (fee = 2%)
20%	1.7	+9.1%	+2153.7%
60%	14.9	+81.6%	+1863.7%
120%	59.6	+326.3%	+884.6%

Finding. At the shipped default fee = 0.02 (= 100 bps half-spread) the fixed mode never inverts within the RV slider range [10%, 140%] — inversion needs RV > 155.4% at fee 2%, or fee < 1.62% at the slider maximum — so the in-app fixed-mode caption ("watch the APR invert") only demonstrates at lower fee settings. The headline contrast (vol-indexed $+3.1/+27.7/+110.8\%$ vs fixed $+24.5/-74.0/-406.4\%$ APR at RV 20/60/120%) is *historical* and belongs to **build 1** (commit `75abb52`, *before* the build-4 gamma-bleed basis fix): not reproducible on build-11 defaults.

7 The formal corpus

Five Lean 4 packages, 130 theorems. Uniformly: no `sorry`/`admit`/`axiom`/`native_decide`/`opaque`/`unsafe`; every theorem's `#print axioms` = {propext, Classical.choice, Quot.sound}; statements byte-identical to what was submitted (prover-side edits are proof bodies and private helpers only); all locally kernel-verified in Lean 4.28.0 (Mathlib `.olean`s from the CI cache — stated, not hidden). Label-flip from *trusted-from-prover* to *verified* is on record as a recommendation; the call is the operator's.

7.1 `BURR2_CORE` — 42 theorems (Aristotle run `26958f2b`)

Theorems about the *exact* production formulas of §2, conditional on nothing but four interface facts about the incomplete beta (carried as a structure field, not an axiom). §1 sheet correspondence and positivity (`uArg_eq_sheet`, `tail_zero`, `G1_pos`, `G1_lt_one`, …); §2 parity (`burr2_parity_of_ne` hypothesis-free off-ATM; `atm_wings_meet`: the peg is *derived* as the unique wing-meeting split; `burr2_parity`); §3 zero-strike anchor (`AL_glue`, $\text{CALL}(-1) = 1$, $\text{PUT}(-1) = 0$); §4 sign and order (19 theorems: kernel in $(0, 1]$ decreasing, wings nonnegative, `CALL_antitone`, `CALL_lipschitz`, `PUT_monotone` — conditional on `TailRep`); §5 non-vacuity witness; §6 apportionment (`harmonic_law`, `shares_sum_one`, `share_eq_agg_ratio`, `share_nonneg`, `apportionment_conserves`). Two substantive corrections the proofs forced on the brief: at-the-money parity *is* the peg (the 1.39×10^{-17} and 2.44×10^{-13} residuals are one phenomenon, not two), and $q_R = W_L/(W_R+W_L)$ is derived, not chosen.

7.2 `BURR2_MIXTURE` — 13 theorems (run `e95f6377`)

Burr-II is *not* closed under maker mixture (`burr2_not_closed_under_mixture`, and the §5 stretch version for scale mixing via a two-atom moment/Cauchy–Schwarz argument), *but* the exact identity

$$\tfrac{1}{2}(\Delta_{\gamma_1} + \Delta_{\gamma_2}) = \Delta_{\bar{\gamma}} \cdot \cosh(\delta\, L_c) \tag{14}$$

bounds the mixture as the same shape times $1 + O(\delta^2)$ — *second* order in the parameter gap, versus the single-lens wall of §4 which is first-order structural. The kernel change is what buys that mildness.

7.3 LOOP_LINK_A (pricing) — 9 theorems (project 20fa5993, task b1e0c962)

The pricing↔aggregation bridge. engine_call_midconvex ($g \geq 1$, via a global supporting line) discharges the standing MidConvex hypothesis for the v28 engine curve; atm_kink_bound in the stronger form $(g+1)(\frac{g+1}{g})^g \geq 2g+1$ for all $g > 0$; engine_book_arb_free / engine_book_parity: a book of makers each with its own exponent g_i is butterfly-arbitrage-free and parity-anchored. And the load-bearing negative half: engine_log_affine + mixture_strict_log_convex $\Longrightarrow$ mixture_not_single_lens — *heterogeneous maker steepness generates a strict smile that no single power-law lens can represent.* This proved obstruction is why the console's makers share the level (§4).

7.4 LOOP_LINK_B (settlement) — 12 theorems (project ce3c2190, task aa5f90f8)

Netting in units is linear and well-defined (units_add, units_smul, cashOne_add); the common doorway neither creates value nor flips signs (cashOne_zero_iff, cashOne_pos_iff); it cannot be timed (exit_timing_irrelevant); and the pair common_doorway_necessary / per_strike_doorway_unbounded_a yields doorway_arbfree_iff_common: one common units→cash factor is *exactly* the arbitrage-free class. "Dollars enter only at the very end" is a theorem.

7.5 sims/v3-maps-lean (base layer) — 54 theorems

The book/aggregation substrate (11 BOOK + 25 MAP + 18 BASIS): agg_parity (weights summing to one), agg_midconvex (nonnegative weights; fixed-across-strikes enforced structurally), book_arb_free (butterfly leg), level_convex_iff_zero_spread_arbfree (a genuine iff, the strongest statement in the file), share_strike_invariant, beta_transport_parallel with the parallel-addition law earned by walk_equiv/walk_cost_equiv (interior case), and the exact vega dichotomy shares_fixed_of_common_transport / common_transport_is_necessary.

8 Open gaps (stated, not hidden)

1. **The Burr-II MidConvex bridge is not proved.** The book-layer arbitrage-freeness chain attaches to the v28 engine power-law curve (engine_call_midconvex) and only *hypothesis-conditionally* to the Burr-II curve this console actually quotes. Named the next Lean obligation.
2. BURR2_CORE §4 sign/order results are conditional on TailRep; the non-vacuity witness is a toy ($a = \gamma = 1$, $I = \mathrm{id}$), not the real incomplete beta.
3. The BURR2_MIXTURE bound is on the midpoint member, near-the-money in scope (residuals of 0.02–0.12% are near-money; 745% excess at γ 0.4 vs 3.0); joint non-closure over all four parameters is unproved.
4. κ (skew) dynamics — the Trade Bands mechanic — are entirely unformalised.
5. **Mislabel risk (base layer):** the exposure readback (Exposure) is cited in v3-maps-lean's README and the loop map but is *absent* from the Lean files (re-verified 2026-08-18); the Portfolio hedge-readback panel has zero Lean backing. Moreover (skeptic finding, 2026-08-18, reported not fixed): its "residual exposure 0.000000" is a hardcoded string over the definitional identity hedge := $-\Delta_{\text{net}}$, and the demo put-leg delta is wrong — the

code uses $-\Delta_{\text{call}} = 0.5358$ at $k = -0.10$ where parity $P = C + k$ forces 0.4642 (manager re-derived: exact).

6. "Arb-free" means *butterfly* only: no unconditional vertical leg, no price bounds ($C \geq$ intrinsic, $C \leq S$); and there is no individual-rationality/adverse-selection theorem for makers under the mixed book (the refraction finding).
7. The economics layer (bleed, h_{fair}, APR, viability) is a model with demo constants and no Lean; the fixed-mode inversion is unreachable at the default fee (see §6). Realtime maker tuning converts a passive-LP product into an active-MM venue — adverse selection, correlated withdrawal, and herding are new, unmodelled risks (operator transcript entry 541).
8. The console is not wired to the production engine; nothing here changes `engine/` or its gates.

A Build history (all 2026-08-14)

Operator-direction fragments below are quoted from the build commit messages; the session's verbatim transcript is missing (corrigendum filed 2026-08-18, `history/operator/CORRIGENDUM_2026-08-18_appbuil`

build	commit	change
1	`75abb52`	Burr-II RFQ book, single file, Railway-ready; parity exact headless. Carries the historical APR headline (§6), pre-basis-fix calibration.
2	`269a6a3`	Deployed to Railway (PORT fix); `BURR2_CORE` + `BURR2_MIXTURE` Lean landed (55 theorems).
3–4	`0e2b10a`, `8dfd02e`	Rebuild on the reference design system; gamma-bleed basis fixed to notional.
6	`caf6c71`	Rung ladder stripped — continuous quoting (operator: "that was the OB"; take the UX elements, not the orderbook mechanics).
8	`07370be`	Bid and ask as first-class separate curves; filled spread band; $\times N$ magnifier with honest caption; build stamp.
9	`7650a2a`	Sampled quote sheet replaced by an any-strike quoter.
10	`7fd1325`	Quote box merged into the chart — hover to quote (operator: "why u need box separately").
11	`ebd774e`	All views activated; aggregate book on Transact; *crossed-book bug found and fixed* — per-maker levels crossed the book (a level-disagreement failure, §4); makers now share the level and differ in depth/spread only; CROSSED — arb detector added.

Twin document: `SPEC_RFQ_MM_CONSOLE.xlsx` in the same directory carries this specification in spreadsheet form, including the theorem-level listing of all 76 RFQ-specific theorems. This document was audited by the project's adversarial skeptic gate (verdict: `notes/skeptic/VERDICT_rfq_spec_docs_2026` its three flags — the build-1 attribution of the historical APR headline, the softening of "forced"/"settled" on the common-level design, and the hardcoded readback residual plus the put-delta finding — are incorporated above.